

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:	G. Bharathi	Reg. No.:	192371079
Date:	13/08/2026	Duration:	60 minutes

I G. Bharathi (192371079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- 1. Study and Analyze the Scenario**
 - Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. Identify Functional and Non-Functional Requirements**
 - List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

 - Introduction
 - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.
- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.
- Check for ambiguity, redundancy, inconsistency, and missing requirements.
- Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS)	SRS is well-structured, complete, professionally organized, and	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or

Documentation (15%)	includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

COLLEGE ADMISSION COUNSELING SYSTEM

Scenario-Based Requirement Analysis and Software Requirement Specification

1. Study and Analyze the Scenario

The College Admission Counseling System is a digital platform that helps students explore colleges, understand admission requirements, receive counseling guidance, submit applications, track application status, and communicate with counselors. Counselors and administrators can manage student profiles, counseling sessions, college information, applications, and reports.

Business Domain

- Higher education admission and counseling
- Student guidance and application management
- College and course information management
- Counselor–student communication

Stakeholders

- Students/Applicants
- Admission Counselors
- College/Department Administrators
- Admission Office Staff
- College/University Information Providers
- Notification Service

Existing Challenges

- Manual counseling records
- Difficulty comparing colleges and courses
- Repeated student queries
- Unclear application status
- Scattered admission information
- Difficulty scheduling counseling sessions
- Time-consuming report preparation

System Objectives

- Centralize admission and counseling information
- Provide personalized counseling support
- Manage student profiles and preferences
- Maintain college, course, eligibility and deadline information
- Schedule counseling sessions
- Track applications and status
- Provide notifications and reminders
- Generate administrative reports

- Reduce manual work and errors

2. Identify Functional and Non-Functional Requirements

Functional Requirements

- FR1 – Student Registration and Login
- FR2 – Student Profile Management
- FR3 – College and Course Search
- FR4 – Eligibility and Admission Information
- FR5 – Counseling Session Scheduling
- FR6 – Counselor Management
- FR7 – Counseling Notes and Recommendations
- FR8 – Application Submission/Tracking
- FR9 – Application Status Updates
- FR10 – Notifications and Reminders
- FR11 – Admin Management
- FR12 – Reports and Analytics
- FR13 – Document Upload and Management
- FR14 – Search and Filtering
- FR15 – Feedback and Support

Non-Functional Requirements

- Performance: Fast search, profile access, and status retrieval.
- Security: Protect student profiles, documents, and login information.
- Reliability: Preserve counseling and application records accurately.
- Usability: Simple interface for students, counselors, and administrators.
- Scalability: Support increasing students, colleges, courses, and applications.
- Maintainability: Modular components and clear separation of responsibilities.
- Availability: System should be accessible during admission periods.
- Privacy: Student information should only be available to authorized users.

3. Identify Actors and Use Cases

Actor	Major Use Cases
Student	Register, Login, Profile, Search Colleges, View Eligibility, Book Counseling, View Recommendations, Track Application, U
Counselor	Login, View Students, Schedule Sessions, Provide Guidance, Add Notes, Recommend Colleges/Courses
Administrator	Manage Users, Colleges, Courses, Counselors, Applications, Reports, Notifications
Admission Office	Update admission information, deadlines and application status
Notification Service	Send reminders and status notifications

4. Software Requirement Specification

Use Case – Book Counseling Session

Actor: Student

Precondition: Student is registered and logged in.

Process: Student selects counseling → chooses counselor/date → system checks availability → student confirms → system stores appointment → notification is sent.

Postcondition: Counseling session is scheduled.

Use Case – Track Admission Application

Actor: Student

Process: Login → open applications → select application → view current status, deadlines and pending documents → receive updates.

Postcondition: Student can monitor application progress.

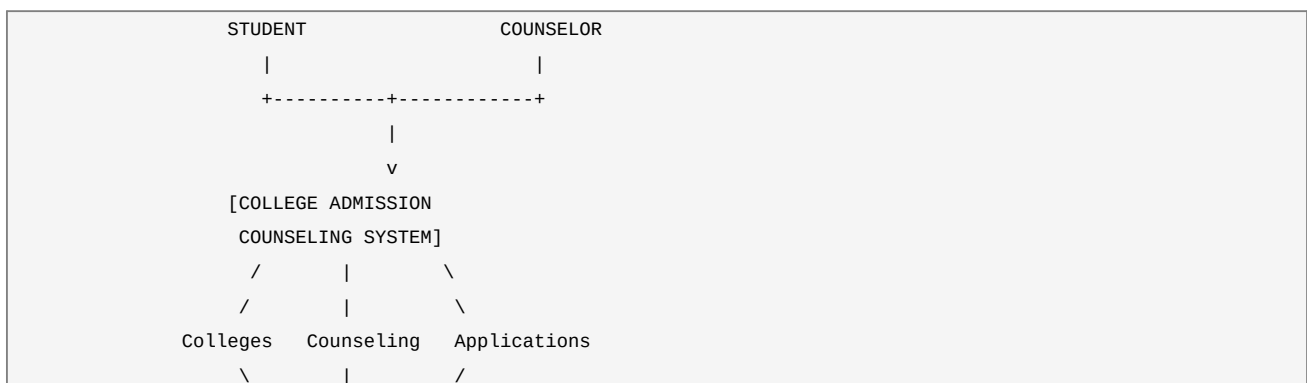
5. Data Requirements

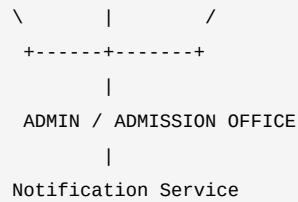
- Student profiles
- Counselor profiles
- College details
- Course/program details
- Eligibility criteria
- Admission deadlines
- Counseling appointments
- Counseling notes and recommendations
- Applications and status history
- Uploaded documents
- Notifications
- Reports and audit logs

6. External Interface Requirements

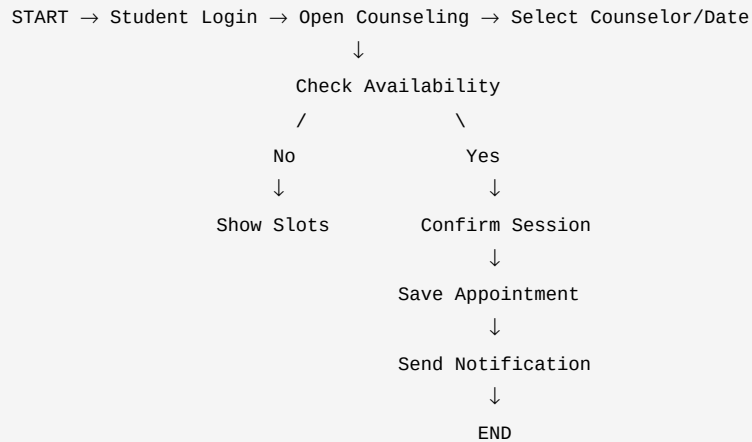
- User Interface: Student dashboard, counselor dashboard, admin dashboard, search, counseling, applications and reports.
- Software Interfaces: Database, authentication service, notification/email service and document storage.
- Communication: Secure HTTPS/API communication.
- Devices: Desktop, laptop, tablet and mobile devices.

7. System Context / Use Case Structure

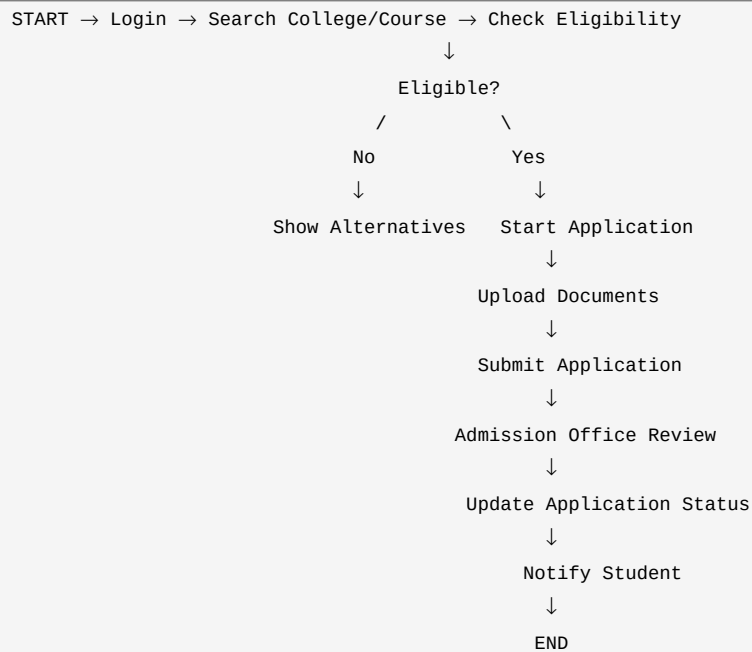




8. Counseling Session Flowchart



9. Admission Application Flowchart



10. Conclusion

The College Admission Counseling System provides a centralized platform for student counseling, college and course exploration, application tracking, counselor interaction, and admission information management. It reduces manual processes, improves transparency, and supports students throughout the admission journey.